

IV.POS.8 D1.C — Bug-Proximity Metric Sta

Corpus: 30 Cat-A DBs (10 V1 + 10 V5 + 10 D1.A decay).

Artifacts from Batches 1–3 under `a4/runs/iv_pos_8/d1c/`.

```
In [ ]: import sys
        from pathlib import Path
        import matplotlib
        matplotlib.use('Agg')
        import matplotlib.pyplot as plt
        import pandas as pd
        import numpy as np
        from IPython.display import Image, display, Markdown

        REPO = Path('/root/arguzz')
        D1C = REPO / 'a4/runs/iv_pos_8/d1c'
        PLOTS = D1C / 'plots'

        tier1 = pd.read_csv(D1C / 'd1c_batch1_tier1_audit.csv')
        metrics = pd.read_csv(D1C / 'd1c_metrics_table.csv')
        corr = pd.read_csv(D1C / 'd1c_correlation_matrix.csv')
        paired = pd.read_csv(D1C / 'd1c_paired_tests.csv')

        def show_png(name):
            path = PLOTS / name
            if path.exists():
                display(Image(filename=str(path)))
```

§1 Tier-1 fire-rate audit (Batch 1)

```
In [ ]: show_png('d1c_batch1_fire_rate_full.png')
        show_png('d1c_batch1_fire_rate_post_local.png')
        summary = tier1.groupby('signal_name').agg(
            fire_rate_full=('fire_rate_full', 'mean'),
            fire_rate_post_local=('fire_rate_post_local', 'mean'),
        )
        display(summary.round(4))
```

§2 Tier-2 metrics table (Batch 2)

```
In [ ]: display(metrics.groupby('corpus')[
            ['cat_a_pro_s5_singleton_failure_rate', 'cat_a_pro_s5_d_loc_p95',
             'cat_a_pro_s5_co_failure_graph_degree_p95']
        ].mean().round(4))

        singleton = paired[paired.metric == 'cat_a_pro_s5_singleton_failure_rate']
        display(singleton[['comparison', 'mean_a', 'mean_b', 'p_value']])
```

§3 Correlation heatmap (Batch 3, V5 paired seed:

```
In [ ]: v5p = corr[(corr.variant == 'V5') & (corr.seed.isin([1234,1235,1236,1237,
pivot = v5p.pivot_table(index='signal_name', columns='channel_name', valu
fig, ax = plt.subplots(figsize=(6, 4))
im = ax.imshow(pivot.values, vmin=-0.4, vmax=0.4, cmap='RdBu_r', aspect='
ax.set_xticks(range(len(pivot.columns)))
ax.set_xticklabels(pivot.columns, rotation=30, ha='right')
ax.set_yticks(range(len(pivot.index)))
ax.set_yticklabels(pivot.index)
ax.set_title('Mean Pearson r (V5 paired seeds, post-local window)')
plt.colorbar(im, ax=ax)
out = PLOTS / 'd1c_batch3_corr_heatmap_v5.png'
fig.savefig(out, dpi=120, bbox_inches='tight')
plt.close(fig)
display(Image(filename=str(out)))
display(pivot.round(3))
```

§4 Shortlist summary

```
In [ ]: shortlist = (D1C / 'd1c_signal_shortlist.md').read_text()
display(Markdown(shortlist.split('## Bucket A')[0] + '## Bucket A (excerp
```

§5 D1.E hand-off pointer

```
In [ ]: display(Markdown((D1C / 'd1e_handoff_L1_signals.md').read_text()[:2000] +
```